

An Examination on Reduction of Displayed Character Shake while Walking in Place with AR Glasses

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ABSTRACT

In recent years, augmented reality (AR) has started to be used in our daily lives. AR glasses are used when walking, which is a normal part of daily life, but walking causes the text displayed on the glasses to shake. This reduces both readability and our attention to what is in front of us, and increases discomfort. We propose a method of fixing the text to take account of shaking while walking to reduce these adverse effects. Experiments revealed the effectiveness of our reduction method and its influence on the distance of the text display.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality; Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Graphical user interfaces

1 INTRODUCTION

In recent years, augmented reality (AR) using smartphones and head mounted displays (HMDs) has been attracting attention. AR is a technology that superimposes visual information on real space, and it is expected to be used in our daily lives. It is known that AR glasses and HMDs allow users to obtain visual information without lowering their eyesight and to pay attention to their surroundings. Active research has been undertaken on fonts that are resistant to screen and text shaking [2] because screen shaking is a problem when walking with AR glasses or HMDs as shown in Figure 1. However, there has been insufficient research on improving the readability of text by reducing the text shaking caused by the screen shaking.

In this study, we investigate the effects of text shaking reduction on text readability, attention to what is ahead, and discomfort. We confirm the advantages of our method as regards reducing text shaking by stabilizing the text.

2 EXPERIMENT

The purpose of this experiment is to determine the effectiveness of our text shaking reduction method on text readability, forward attention, and discomfort. In this experiment, we conduct a subject experiment by changing the distances of the text on the HMD, and comparing the conventional and reduction methods.

2.1 Experiment Design and Tasks

The number of characters to be displayed was set at seven, and they were randomly selected from hiragana (Japanese characters) consisting of 50 syllables. It is possible for a pedestrian to read this number of characters while walking 5 m. The font was MS Gothic, which is a standard font in Windows. We used white (#FFFFFF) as the font color. Since the background fluctuates in AR, a pilot test

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Figure 1: Text shaking displayed on an HMD

was conducted with five subjects to determine the font color that is least affected by the background color. As a result, we used a white font whatever the background.

The text was positioned in the center. According to a related study [3], the readability of the center was significantly greater when the screen was divided into 9 sections.

Three text distances were used: 1) A long distance of 1.5 m to display a 4.0 cm character based on the related study, 2) A short distance of 0.45 m to display a 1.2 cm character based on Microsoft's Mixed Reality typography, 3) A medium distance of 0.75 m to display a 2.6 cm character. The medium distance is intermediate between the long and short distances. All three use the same viewing angle of 1.53 degrees, so the size of the visible font is same.

Two display methods were prepared. One is the conventional method, which does not perform any processing, and the other is the reduction method, which stabilizes the text in the AR space.

In our daily lives, it is possible to read when the head is moving and the text is stationary. However, it is difficult to read when the text is moving and the head is stationary. This is influenced by an ocular reflex called the vestibulo-ocular reflex (VOR) [1]. The VOR can achieve visual stability because we learn the shaking while we walk. Figure 2(i) shows that the head is moving and the text is stationary, which is learned in daily life. However, in the case of the conventional method shown in Figure 2(ii), the text follows the movement of the head, which means that both the head and the text move, making it even more difficult to read. Therefore, in the reduction method shown in Figure 2(iii), a technique was introduced in which the head moves and the text appears to stop, just as in daily life (Figure 2(i)). The method is as follows: First, the head position is recorded before the start of walking. After that, the difference between the recorded position and the current position of the head while walking is calculated. The difference is used to move the text up and down to display it at a fixed location.

A virtual cone appears 2 m ahead of the subject to check whether the subject is paying attention to what is in front without being absorbed in reading the text.

2.2 Procedure

Subjects performed steps 1 through 4 below.

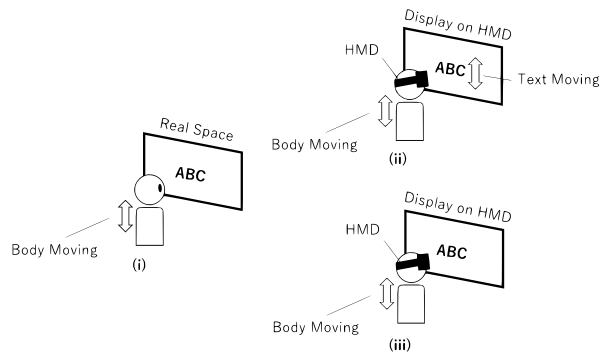


Figure 2: (i) Reading the text while moving in real space, (ii) Reading the text with the conventional method of AR, and (iii) Reading the text with the reduction method of AR

1. Wear an HMD and run on the spot.
2. Press the button on the controller to display the AR text.
3. When a virtual cone appears in front of the subject (at the right/ center/ left), read the text aloud while paying attention to what is ahead.
4. Evaluate Q1 to Q3 on a 7-point scale (very strongly disagree (-3)/agree (3), strongly disagree (-2)/agree (2), slightly disagree (-1)/agree (1), neutral (0)) and answer Q4.

The contents of Q1 to Q3 are shown below.

- Q1 Did you feel that the text was easy to read?
 Q2 Did you feel that it was possible to direct your attention forward?
 Q3 Did you feel uncomfortable with the shaking of the text caused by walking?

To check whether they were paying attention to what was in front, we asked them Q4 "Where was the position of the cone?". The subjects chose the answer left, center, or right.

The subjects repeatedly practiced this sequence of actions until they became familiar with it. The subjects then performed under all six conditions: three different text distances and two different display methods. The order of the conditions was randomized for each subject. There were 22 subjects (1 female and 21 male university students).

3 RESULTS AND DISCUSSION

A total of 132 data were analyzed (6 conditions x 22 subjects), and since the percentage of wrong answers in Q4 amounted to less than 2.0% of the total, it was considered that the subjects fulfilled the condition of reading the text while retaining attention to what was ahead of them.

Figure 3 shows the results of Q1-Q3, the mean and standard error, for the display methods (the conventional and reduction methods). In Q1, readability was significantly improved from negative to positive by using the reduction method (t -test, $t(65)=-3.04$, $p<.01$). As regards the forward attention in Q2, both were possible, but the reduction method significantly improved the results (t -test, $t(65)=-2.10$, $p<.05$). In Q3, discomfort was felt in both cases, but it was significantly reduced when using the reduction method (t -test, $t(65)=2.85$, $p<.01$). Therefore, it was found that the use of the reduction method increased the readability of the text and the attention to the front. And it reduced the discomfort of shaking. This result not only confirms the VOR study, but also shows the effectiveness of the reduction method in AR.

Next, we compare the effects of the different methods on the distances of the text (far, medium, and short distances). First, Figure 4

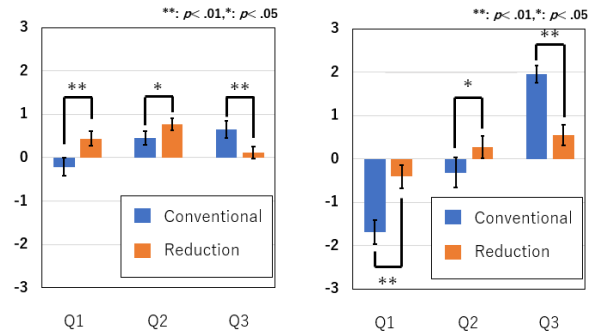


Figure 3: Results of Q1-Q3 by the display methods (Conventional, Reduction)

Figure 4: Results of Q1-Q3 by the display methods (Conventional, Reduction) at a short distance

shows the results of the mean and standard error between the methods at the short distance. In Q1, the readability was negative for both methods, but significantly improved with the reduction method (t -test, $t(21)=-3.71$, $p<.01$). Q2 was negative for the conventional method, but positive for the reduction method, and the forward attention was significantly improved with the reduction method (t -test, $t(21)=-2.14$, $p<.05$). In Q3, the discomfort was significantly reduced by using the reduction method (t -test, $t(21)=-4.83$, $p<.01$). Then, as regards the results for the methods at the medium distance, the reduction method was better than the conventional method in Q1 (t -test, $t(21)=-1.85$, $p<.10$), and there was no significant difference in Q2 and Q3. And finally, there was no significant difference with the long distance in Q1, Q2 and Q3. Therefore, it was found that the shorter distance was, the more the reduction method could improve the readability and direct the attention to the front. It was also found that the discomfort was reduced for the reduction method. The short distance is 0.45 m in front of HMDs, which is the distance employed in daily life when using AR. Thus, the reduction method is better in terms of practical use.

4 CONCLUSION

In this study, we proposed a method for reducing reading difficulty, inattention to the region in front, and discomfort due to the text shaking when walking while wearing AR glasses.

From the experimental results, we found that implementation of the reduction method increased both readability and attention to the region ahead. In addition, discomfort was reduced with this method. Moreover, it was confirmed that the reduction method was more effective when the text was displayed at a short distance.

In the future, although the effectiveness of the reduction method at a short distance was confirmed, the range of distances at which the reduction method is effective remains to be examined.

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